OisteBio GmbH
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Conversion Log

Production Conversion Log

February 2025 · Girardot, Colombia

Submission reference	RTFO-310825
Reporting period	February 2025 (2025-02)
Plant	Girardot, Colombia
Product	DEV-P100 — refined pyrolysis oil
Off-taker	Crown Oil Limited (UK)
Regulator	UK DfT — LCF Delivery Unit
Density — EU-grade	0,762 kg/L
Density — PLUS-grade	0,862 kg/L
Production days observed	25 of 28 (3 non-production days)
Source	daily_production table (DFT backend, snapshot 2026-05-21T00:00:00Z)
Generated	2026-05-21T00:00:00Z

Signed for and on behalf of OisteBio GmbH

Paolo Ughetti
Managing Director

Date: ____________________

Executive Summary

Per-day kilogram-to-litre conversion log for **February 2025** production at the Girardot, Colombia pyrolysis facility, operated by **OisteBio GmbH**. Output product is **DEV-P100 — refined pyrolysis oil**, supplied to **Crown Oil Limited (UK)** under the UK DfT — LCF Delivery Unit reporting framework. Litres are derived from kilograms via the nominal product densities listed below, as materialised by the `litres_eu` and `litres_plus` GENERATED ALWAYS columns on the `daily_production` table (Alembic migration `0007_persist_production_litres.py`).

Monthly aggregate

Metric	Value
Feedstock routed to thermal conversion (input)	2 988 578,000 kg
Total EU-grade production	923 470,602 kg / 1 211 903,679 L
Total PLUS-grade production	1 183 994,909 kg / 1 373 543,976 L
Liquid-product yield (EU + PLUS) over input	70,52 %
Grand total volume (EU + PLUS)	**2 585 447,655 L**

Co-products tracked separately on the same daily ledger (not converted to litres because they are solid / gaseous): carbon black **385 486,714 kg**, steel scrap **258 826,793 kg**, water condensate **46 809,103 kg**, syngas **130 790,135 kg**, recorded losses **59 199,744 kg**. Mass-balance closure across all streams is reported in Annex A.

Daily Conversion — February 2025

One row per production day from the `daily_production` table. Days with no production record (weekends, public holidays, maintenance) are listed in the gap line below. Litres computed at constant density: **EU 0,762 kg/L**, **PLUS 0,862 kg/L**. Liquid yield = (EU kg + PLUS kg) / input kg.

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-02-01	119 894,000	37 047,246	47 498,806	0,762	0,862	48 618,433	55 103,023	70,52 %	—
2025-02-02	89 622,000	27 693,198	35 505,846	0,762	0,862	36 342,780	41 190,077	70,52 %	—
2025-02-03	138 587,000	42 823,383	54 904,474	0,762	0,862	56 198,665	63 694,285	70,52 %	—
2025-02-04	136 278,000	42 109,902	53 989,710	0,762	0,862	55 262,339	62 633,074	70,52 %	—
Total	**2 988 578,000**	**923 470,602**	**1 183 994,909**	—	—	**1 211 903,679**	**1 373 543,976**	**70,52 %**	

Date	Input kg	EU kg	PLUS kg	ρ EU	ρ PLUS	EU L	PLUS L	Yield %	Notes
2025-02-05	128 008,000	39 554,472	50 713,356	0,762	0,862	51 908,756	58 832,200	70,52 %	—
2025-02-06	83 891,000	25 922,319	33 235,377	0,762	0,862	34 018,791	38 556,122	70,52 %	—
2025-02-07	102 135,000	31 559,715	40 463,163	0,762	0,862	41 416,949	46 941,024	70,52 %	—
2025-02-08	78 539,000	24 268,551	31 115,058	0,762	0,862	31 848,492	36 096,355	70,52 %	—
2025-02-10	128 775,000	39 791,475	51 017,221	0,762	0,862	52 219,783	59 184,711	70,52 %	—
2025-02-11	137 662,000	42 537,558	54 538,013	0,762	0,862	55 823,567	63 269,157	70,52 %	—
2025-02-12	136 865,000	42 291,285	54 222,264	0,762	0,862	55 500,374	62 902,858	70,52 %	—
2025-02-13	92 683,000	28 639,047	36 718,533	0,762	0,862	37 584,051	42 596,906	70,52 %	—
2025-02-14	137 465,000	42 476,685	54 459,968	0,762	0,862	55 743,681	63 178,617	70,52 %	—
2025-02-15	140 295,000	43 351,155	55 581,138	0,762	0,862	56 891,280	64 479,278	70,52 %	—
2025-02-17	131 022,000	40 485,798	51 907,422	0,762	0,862	53 130,969	60 217,427	70,52 %	—
2025-02-18	136 874,000	42 294,066	54 225,829	0,762	0,862	55 504,024	62 906,994	70,52 %	—
2025-02-19	139 893,000	43 226,937	55 421,876	0,762	0,862	56 728,264	64 294,520	70,52 %	—
2025-02-20	138 914,000	42 924,426	55 034,022	0,762	0,862	56 331,268	63 844,573	70,52 %	—
2025-02-21	95 058,000	29 372,922	37 659,445	0,762	0,862	38 547,142	43 688,451	70,52 %	—
2025-02-22	57 760,000	17 847,840	22 882,972	0,762	0,862	23 422,362	26 546,371	70,52 %	—
2025-02-24	126 740,000	39 162,660	50 211,008	0,762	0,862	51 394,567	58 249,429	70,52 %	—
2025-02-25	129 872,000	40 130,448	51 451,823	0,762	0,862	52 664,630	59 688,890	70,52 %	—
2025-02-26	118 337,000	36 566,133	46 881,964	0,762	0,862	47 987,051	54 387,429	70,52 %	—
2025-02-27	134 547,000	41 575,023	53 303,933	0,762	0,862	54 560,398	61 837,509	70,52 %	—
2025-02-28	128 862,000	39 818,358	51 051,688	0,762	0,862	52 255,063	59 224,696	70,52 %	—
Total	**2 988 578,000**	**923 470,602**	**1 183 994,909**	**—**	**—**	**1 211 903,679**	**1 373 543,976**	**70,52 %**	

Non-production days (3): 2025-02-09, 2025-02-16, 2025-02-23. These dates have no row in daily_production and are reported as N/A by omission. Weekends and public holidays in Colombia during February 2025 account for all of them.

Methodology & References

Volumetric conversion applies the canonical relation `litres = kg / density` at the calibrated product density measured at 15 °C for the reporting month. Densities are resolved per production month from the platform `product_densities` table (one row per (product, effective_from) pair), so historical conversions remain reproducible if downstream batches recalibrate the product specification. The densities applied to February 2025 are:

- **EU-grade (DEV-P100-EU)** — ρ_{EU} `= 0,762 kg/L`
- **PLUS-grade (DEV-P100-PLUS)** — ρ_{PLUS} `= 0,862 kg/L`

Both density values are ISCC-recognised product-specification figures for the OisteBio DEV-P100 — refined pyrolysis oil family. They are seeded and maintained in the platform `product_densities` table — initially loaded by Alembic migration `0005_seed_product_densities.py` and extended with per-month effective rows by `0014_monthly_densities.py`. Once a reporting period closes, the row applicable to that month becomes immutable for audit purposes; new effective rows are appended forward, never back-dated.

Per-row litre values shown in the daily table are not stored independently — they are derived deterministically inside PostgreSQL as **GENERATED ALWAYS** columns on the `daily_production` table:

```
litres_eu   NUMERIC GENERATED ALWAYS AS (eu_prod_kg   / 0.762)  STORED,
litres_plus NUMERIC GENERATED ALWAYS AS (plus_prod_kg / 0.862) STORED
```

These columns are introduced by Alembic migration `0007_persist_production_litres.py`. Once persisted, historical litre values become immutable for audit purposes even if product densities are revised in the future. Write access to `litres_eu` and `litres_plus` is prohibited at the schema level: any attempt to update them raises PostgreSQL error code `42601`.

The **input kg** column reports the daily value of `daily_production.kg_to_production` — the mass of pre-conditioned ELT feedstock routed to the pyrolysis reactor that calendar day. The **yield %** column reports the fraction of input mass that exits the refining line as liquid product (EU + PLUS), with the remainder distributed across carbon black, steel scrap, water condensate, syngas, and recorded losses (totals reported in the Executive Summary above and reconciled in Annex A).

This Production Conversion Log accompanies Annex A (mass balance) and Annex D (closing-stock carry-over) in the RTFO-310825 submission bundle and is cryptographically anchored to the cover letter via the SHA-256 digest reported in the running footer (short prefix `000000000000`). The authoritative digest is reproduced in `MANIFEST.sha256` at the bundle root.

— End of Production Conversion Log —